Eduardo Ramos Mendoza

Computer Systems Engineer

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Profile

Systems engineer specializing in high-performance and LLM-agent infrastructure. Experience with processor architecture, FPGA design, network programming, and full-stack development.

Education

Currently enrolled — **Computer Systems Engineering**, *Escuela Superior de Cómputo, Instituto Politécnico Nacional*, CDMX

Selected Projects

AI agents — Phyriager - embodied LLM agent in Minecraft: Java/Fabric mod ↔ JNI ↔ shared-memory bridge to an external C++ brain; local-LLM planner with VLM vision. Source: GitHub repository

Real-time frame generator — PhyriadFG - real-time frame pipeline: external capture → optical-flow frame generation → upscaling → paced presentation, routed across multiple GPUs by measured capability. Source: GitHub repository

Runtime tuning — Phynned - runtime optimizer that pins game workloads to the strongest cores of asymmetric CPUs (AMD X3D V-Cache, Intel hybrid). Source: GitHub repository

Hardware — PHarvard - 32-bit modified-Harvard CPU in structural VHDL: MIPS-format ISA. Source: GitHub repository

Web frontend — climx - live weather site covering all 2,463 Mexican municipalities on official SMN/CONAGUA data: a scheduled GitHub Actions pipeline. Live: swately.github.io/climx. Source: GitHub repository

Professional Experience

2020–Present — **Server Administrator & Systems Developer**, *Freelance / Personal Projects*, CDMX

- Set up and administered multiplayer game servers: availability, latency optimization, and user management.
- Administered Linux containers (LXC) on two Proxmox VE hosts: provisioning, per-container CPU/RAM allocation, systemd services, and scheduled backups with retention policies.
- Performed preventive and corrective maintenance of computers and servers (Windows and Linux).

Technical Skills

Programming skills — Programming logic, object-oriented programming, component-oriented programming, Clock-Driven Programming

Systems & HPC — C++23, C, CUDA, Vulkan compute, TCP/UDP sockets, CMake/Ninja, benchmark methodology

AI / LLM systems — Multi-agent LLM orchestration and optimization, local LLM serving (Ollama).

Hardware — FPGA programming (VHDL, Verilog), digital logic design, circuit analysis, microprocessor programming, DC/servo motor control

Web & Mobile — React, React Native, PHP (REST APIs, MySQL), component-based design

Tools — Git + GitHub Actions CI, Linux, Windows, Wireshark, LaTeX, Lattice Diamond

Python & Data Libraries — pandas, NumPy, TensorFlow, scikit-learn, Prophet, Matplotlib / Seaborn, data cleaning and preprocessing pipelines, exploratory data analysis, model training and evaluation

Languages

Spanish — Native

English — Intermediate (B1)

Russian — Beginner (A1)